A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 8 nights **1** The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards – extra samples do not change it significantly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size **1** Sampling at different conditions / times would make results not comparable **1** Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1 Accuracy:** how close a recorded result is to the real value **1 Precision:** the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value • **precision** → readings close together / fine resolution	2
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level **1** Standardised techniques ensure results collected at different times, places or by different people are comparable **1** Other variables are controlled or monitored so their effect can be assessed – allows the study to be repeated / checked (reliability) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **correlation** → a relationship in which the two variables change together • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned **1** HCFCs to be phased out (by 2030) – some essential uses (eg halon fire extinguishers in aircraft) still permitted **1** A fund to help (developing) countries implement the Protocol – legally binding targets (unlike the 1985 Vienna Convention) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
02.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Aerosols:** pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk **2 Refrigerants:** HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** CFCs are drained from the appliance and incinerated / broken down at high temperature **1** Products are CO2 and acidic gases (HCl, HF) which are neutralised with crushed lime **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	1 mark for the point below. **1** CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years – some ODSs (eg dichloromethane) are not banned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** High mean wind velocities that are consistent in strength and direction **1** A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber – the air forced in and out drives a turbine – surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine) – overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines) – oscillating wave surge converter (Oyster) **OR** point absorber (floating buoy moving relative to a fixed base turns a generator) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Equipment must withstand storms and corrosion **1** Difficult to anchor off stormy coasts / in deep water **1** Expensive to transmit electricity from isolated coasts to consumers **1** Competition with established technologies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
03.4	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Barrage:** huge quantities of construction materials – reduces tidal range so intertidal mudflats used by wading birds are lost – changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity – pollutants concentrate in the lagoon – blocks migratory fish and shipping **up to 2 Lagoon:** similar but smaller scale – only part of the estuary enclosed so the most sensitive areas and main channel can be avoided **up to 2 In-stream turbines:** do not block the tidal flow so little impact on range or sediment – but noise affects cetaceans and many dispersed turbines could have a larger total impact – harness less energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) **1** Most phosphorus compounds have low solubility so little is available in soil water **1** Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil **1** Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake • **symbiotic** → a close relationship in which both species benefit	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) **1** Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment **1** Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates) – harvesting removes phosphorus from farmland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
05.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock **1** Breed crops that absorb phosphate more efficiently **1** Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) **1** Recover phosphate from sewage effluent by iron(III) sulfate precipitation – reduce soil erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (68 000 - 7 000) / 68 000 x 100 **1** = 89.7% (accept 90%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued **1** Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 4 points below. **1** Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress – pollution by pesticides, fertilisers and industrial waste concentrated in the lake **1** Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) **1** Loss of wetland habitats and wildlife – local climate more extreme (loss of the moderating effect of the water body) **1** Loss of livelihoods – conflict between the countries sharing the rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **salinisation** → salt building up in the soil / soil becoming salty **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Upstream abstraction for irrigation reduces the flow available downstream **1** Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream – growing populations increase demand **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
07.1	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Discharge of tank-washing water from tankers **1** Oil rig / drilling accidents / leakage during drilling **1** Pipeline leaks **1** Leakage from storage tanks / refinery spills **1** Waste lubricating oil from vehicles / industry washed into drains **1** Other ship accidents (fuel oil) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia **1** Feathers lose buoyancy / waterproofing so heavily oiled birds drown **1** Birds swallow toxic oil when preening – time spent cleaning feathers reduces time for feeding themselves / their young **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen **1** Toxic components (benzene, xylene) dissolve and poison plankton near the surface **1** Oil blocks light so algae / phytoplankton cannot photosynthesise **1** Oil masks odours / narcotises so animals cannot find mates or food by smell **1** Shore organisms (molluscs, crustaceans) are covered and asphyxiated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Dispersal / dilution by currents and waves is prevented so oil stays concentrated – in open water most spills disperse in days or weeks **1** Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 15 - (-18) = 33 °C **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** Visible light passes easily through the atmosphere and is absorbed by the surface, which warms **1** The warm surface emits (long-wave / far) infrared radiation **1** Greenhouse gases (water vapour, CO_2, methane) absorb infrared better than the average for the atmosphere and convert it to heat **1** The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	4
08.3	1 mark for the point below. **1** Any gas that is better at absorbing infrared radiation than the average for the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	1
08.4	**1 mark each** 1 mark for each of the 2 points below. **1 Arriving:** mainly ultraviolet, visible light and near infrared (short wavelength) – leaving: mainly long-wavelength (far) infrared **1** Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
08.5	**1 mark each** 1 mark for each of the 2 points below. **1** The energy and the processes it drives control climate, ocean currents and the hydrological cycle **1** These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) **1** It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
09.2	**Any 4 of 8** 1 mark each for any 4 of the 8 points below, in any order, maximum 4. **1** Physical state of the effluent (solution / solid) **1** Ocean currents / river flow **1** Effect of pH and oxygen on solubility **1** Bioaccumulation and food-chain concentration **1** Food sources and consumption by local people (eg shellfish, seaweed) **1** Half-lives of the isotopes **1** Meteorological conditions for any gaseous release **1** Nature of the environment - geology, vegetation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	4
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring focuses on the sites most at risk of contamination **1** No need to sample sites that were never likely to be polluted – sampling checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** It assesses the risk to the people most exposed because of their lifestyle before any health effects occur **1** If their risk is acceptably low everyone else has a lower risk – if unacceptable, emissions are reduced to prevent future exposure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **State of matter:** solids deposit near the source, gases disperse widely in the atmosphere – density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • **Persistence / half-life:** persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution – non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk – LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • **Reactivity:** reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium – can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose – exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO2; smoke + SO2) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • **Evaluation:** properties of new chemicals can be predicted from similar ones (precautionary principle)	9

Do not write outside the box

Question	Marking guidance	Marks
	– understanding allows prevention rather than clean-up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **Persistence** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	
	Model answer - Level 3 (9 marks)	
	A pollutant's properties largely determine where it goes and how much harm it does, so they can be used both to predict its behaviour and to choose a control. State of matter and density govern dispersal: solids deposit near the source, gases spread widely, and dense gases such as hydrogen cyanide at Bhopal hug the ground. Persistence decides whether dilution is enough: non-persistent wastes like sewage can be discharged where they break down, but organochlorines, CFCs, heavy metals and long-lived radioisotopes travel far and accumulate, so they need containment, bans or restrictions instead. Toxicity, specificity and mode of action, quantified by LD50 and used to set acceptable doses, identify which species are at risk. Solubility explains why nitrate leaches and is controlled by nitrate vulnerable zones, while liposolubility explains why DDT and methyl mercury bioaccumulate and biomagnify along food chains. Reactive pollutants may degrade quickly or form secondary pollutants - ozone, acid rain, PANs - so controlling the primary emissions with catalytic converters or flue-gas desulfurisation prevents the secondary ones. Adsorption on clay immobilises metals and caesium and is exploited deliberately in activated-carbon and polymer filters. Mutagens and carcinogens have no safe dose, so exposure is kept As Low As Reasonably Achievable, and synergism between ozone and sulfur dioxide, or smoke and sulfur dioxide, means pollutants must be assessed together.	
	Properties interact with the environment - inversions, currents, pH, oxygen - which is why Critical Pathway Analysis combines both to predict where pollutants concentrate and to focus monitoring. The greatest value of this understanding is preventive: the properties of a new chemical can be predicted from similar ones under the precautionary principle, allowing control before release rather than clean-up afterwards.	
	Weaker answer - Level 1 (2 marks)	
	Pollutants have properties like toxicity, persistence and solubility. Toxic pollutants are dangerous. Persistent pollutants like DDT last a long time in the environment and build up in food chains. Soluble pollutants dissolve in water and get into rivers. Knowing these properties helps scientists to predict what pollutants will do and to control them, for example by banning DDT.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 1 defines three properties in isolation; Level 3 pairs every property with a prediction *and* a control strategy, which is what the question asks.	
	• Level 3 uses a wide range of accurate examples (Bhopal, nitrate zones, methyl mercury, PANs, caesium adsorption); Level 1 uses DDT twice.	
	• The final paragraph links properties to environmental conditions (CPA) and to the precautionary principle - synoptic evaluation that distinguishes the top band.	
10.2	1 mark for the point below. **1** The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • **Renewable features:** intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal) – low energy density – location – low pollution but manufacturing impacts • Technologies and developments for each resource with examples – new technologies improving usability • **Storage and grid management:** pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving – the role of electricity as a secondary fuel • **Economics and politics:** development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution – sustainability principles - natural systems run on low-density renewable energy • **Judgement:** electricity largely yes	25
	– heat, aviation, shipping and high-temperature industry harder – transition period with nuclear / CCS *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Features of fertile soil:** water, nutrients, air, organic matter, pH, biota, texture, structure, depth – slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • **Causes of degradation:** vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • **Types of erosion:** rain splash, runoff, wind, slumping – consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • **Management:** long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy	25
	• Evaluation and judgement on which strategies are most effective in different environments *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	

Do not write outside the box